# SmartEdu Analytics: Live Commands Cheatsheet

Step-by-step verified terminal commands for each presenter during evaluation

## 0. ONE-TIME CLOUDERA VM SETUP (Run Once Before Presentation)

```
cd ~
wget -O smartedu.tar.gz http://192.168.56.1:8889/smartedu.tar.gz || wget -O smartedu.tar.gz
http://10.0.2.2:8889/smartedu.tar.gz
tar -xzvf smartedu.tar.gz -C ~
cd ~/SmartEduAnalytics
chmod +x scripts/phase2/*.sh scripts/phase3/*.sh scripts/phase3/*.py scripts/phase4/*.sh run_all_cloudera.sh
```

## SPEAKER 1: HDFS Storage & Health Inspection (Cloudera VM Terminal)

**Terminal Environment:** Cloudera QuickStart VM 5.13.0 (User: cloudera)

**1. Navigate to Project Directory:**

```
cd /home/cloudera/SmartEduAnalytics
```

**2. Ingest Dataset into HDFS:**

```
bash scripts/phase2/upload_dataset_to_hdfs.sh . -f
```

**3. List All HDFS Directories & Files Recursively:**

```
hdfs dfs -ls -R /smartedu
```

**4. Verify HDFS Health & 128 MB Block Allocation:**

```
hdfs fsck /smartedu/processed/preprocessed_student_data.csv -files -blocks
```

**5. Preview First 5 Rows of HDFS Preprocessed Data:**

```
hdfs dfs -cat /smartedu/processed/preprocessed_student_data.csv | head -n 5
```

**6. Open NameNode Web Interface in Browser:**

```
Firefox URL: http://localhost:50070 (Utilities -> Browse the file system -> /smartedu/processed)
```

## SPEAKER 2: MapReduce Streaming & Apache Hive (Cloudera VM Terminal)

**Terminal Environment:** Cloudera QuickStart VM (Hadoop 2.6.0, YARN, Hive 1.1.0)

**1. Execute Distributed MapReduce Job (Fail-Safe with auto-clean):**

```
hdfs dfs -rm -r -f /smartedu/analytics/mapreduce/demo_avg_job 2>/dev/null
hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
-input /smartedu/processed/preprocessed_student_data.csv \
-output /smartedu/analytics/mapreduce/demo_avg_job \
-mapper "python mapper_average.py" \
-reducer "python reducer_average.py" \
-file scripts/phase3/mapper_average.py \
-file scripts/phase3/reducer_average.py
```

**2. View YARN ResourceManager UI (Shows SUCCEEDED status):**

```
Firefox URL: http://localhost:8088
```

**3. Read MapReduce Output Partitions from HDFS:**

```
hdfs dfs -cat /smartedu/analytics/mapreduce/demo_avg_job/part-* | head -n 10
```

**4. Initialize Hive Data Warehouse Schema:**

```
bash scripts/phase4/create_hive_database.sh
```

**5. Interactive Apache Hive CLI Queries (HiveQL):**

```
hive
USE smartedu;
SHOW TABLES;
DESCRIBE students_ext;
SELECT course_id, COUNT(*) AS total_students, ROUND(AVG(annual_grade_avg), 2) AS avg_gpa FROM students_ext
GROUP BY course_id ORDER BY total_students DESC LIMIT 5;
exit;
```

## SPEAKER 3: Web Dashboard & System Integration (Windows Host)

**Environment:** Windows PowerShell Terminal & Web Browser

**1. Launch FastAPI Server & Web Application:**

```
cd d:\bdampr\SmartEduAnalytics
d:\bdampr\.venv\Scripts\python.exe run_all.py --server-only
```

**2. Open Interactive Web Dashboard in Browser:**

```
Browser URL: http://127.0.0.1:8000/ (Admin: admin / smartedu2026)
```

**3. Walkthrough Sequence in Dashboard:**

```
- Overview Tab: Point to 4,424 Total Students, 10.44 GPA, 1,033 High Risk (23.3%), and Doughnut Chart
- Student Roster Tab: Search 'STD_11688' -> Click 'Profile' modal to show Semester 1 vs 2 grades &
interventions
- Dropout Risk ML Tab: Point to Gradient Boosting champion benchmark (93.06% ROC-AUC, 88.70% Acc)
- Top Header: Click 'Export Predictions CSV' to download complete scored dataset
```

**4. Automated Backend API Verification Suite:**

```
d:\bdampr\.venv\Scripts\python.exe phase10_dashboard\backend\test_api.py
```